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Centre Number

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Student Number

2021 YEAR 12

Advanced Mathematics

Task 1 Validation Task

9/12/21

Q	Marks
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

General

Instructions:

- Reading time – 5 minutes
- Working time – 35 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

Total Marks: 28

This question paper must not be removed from the examination room.

This Validation task constitutes 60% of the Assignment.

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

Solve $3^x = 25$, giving your answer to 2 significant figures. 2

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Question 2 (2 marks)

Solve $3 \log_2(10) + \log_2(x) = 6$. Express answer in simplest form. 2

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Question 3 (3 marks)

Solve $2^{x^2} = 8^{\frac{2}{3}x+1}$. 3

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Question 4 (3 marks)

Differentiate the following, simplifying where necessary.

a) $y = 2e^{4x-3}$

1

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b) $y = (2 - e^{-x^2})^6$

2

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Question 5 (2 marks)

If $\log_x(6) = a$ and $\log_x(3) = b$, find $\log_x(8)$ in terms of a and b .

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Question 6 (5 marks)

Radioactive decay is a process which causes a substance to break down over time. The half-life of a substance refers to the amount of time it takes for the substance to decay to half its original mass.

Consider the formula for the radioactive decay of the substance carbon-14,

$$C = C_0 e^{-kt},$$

where C_0 and k are constants, t is time in years, and C is the amount of carbon-14 measured in grams.

- a) Find the value of C_0 if there is initially 1000 grams of carbon-14. 1

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- b) The half-life of carbon-14 is known to be 5730 years. 3

Use this to show that $k = \frac{\ln(2)}{5730}$.

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- c) Hence, find the amount of carbon-14, to the nearest gram, after 10,000 years. 1

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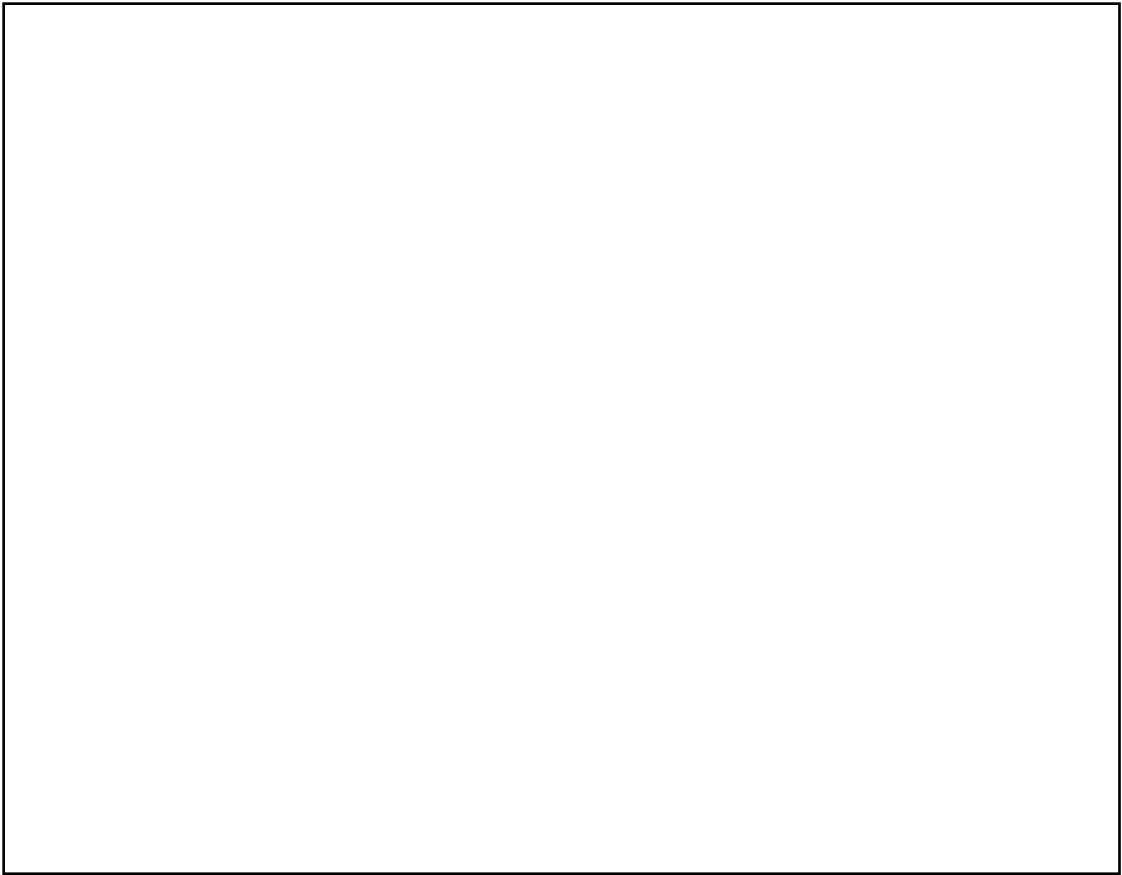
Question 7 (3 marks)

- (a) Sketch the graph of the following exponential function in the box below.
Label x and y intercepts and the asymptote.

3

$$y = e^{-x} - 2$$

Note: The box suggests the size of the expected graph



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Question 8 (6 marks)

Sound intensity level (L) measured in decibels (dB) is a logarithmic measure given as:

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

The reference level of the intensity of sound (I_0) is 10^{-12} watts/m².

- (a) The table below shows the average sound intensity levels (L) and the intensity of the sound (I) of four instruments of a symphony orchestra.

2

Musical Instrument	Average sound intensity level, L (to nearest dB)	Intensity of sound, I (watts/m ²) to 7 decimal places
Cello	87	(i)
Violin	93	0.0019953
Trombone	100	0.0100000
Timpani and bass drum rolls	(ii)	0.0501187

Determine the missing values of (i) and (ii).

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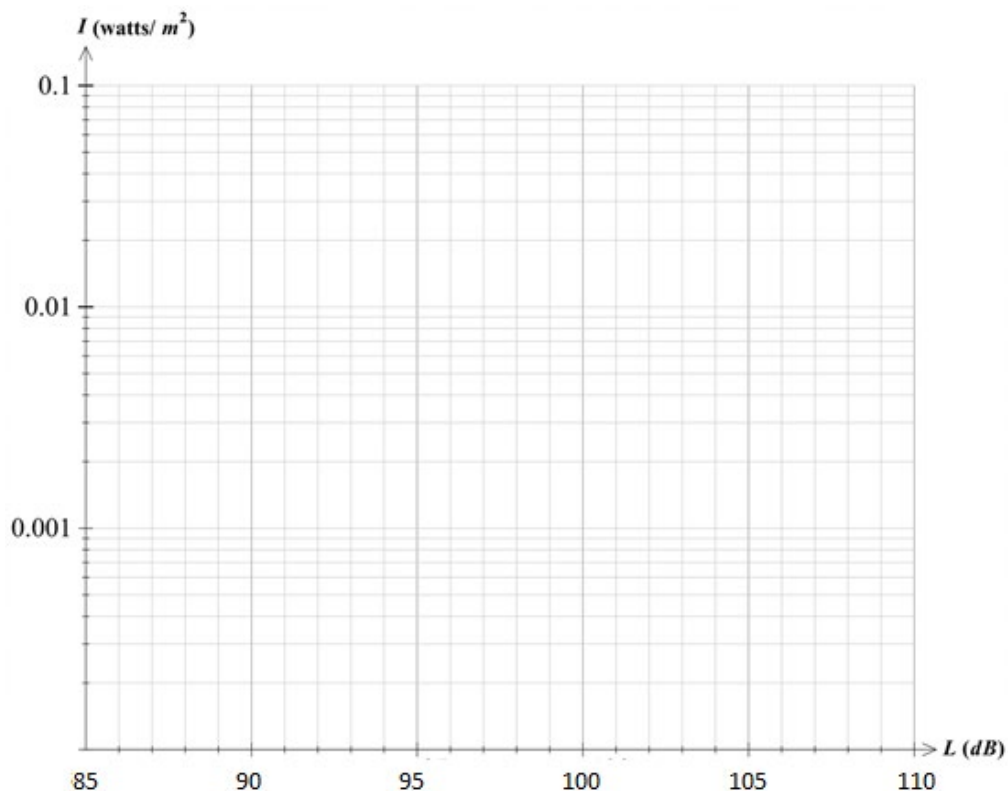
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- (b) Plot the intensity of sound (I) against the average sound intensity levels (L) for the **four instruments listed in part (a)** on the semi-logarithm graph below. **2**
Connect the points with a line.



- (c) Using the graph determine the average sound intensity level(dB) of the sound from a musical instrument that has an intensity of 0.006 watts/m^2 . **1**

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- (d) Explain why a semi-logarithm grid is useful for this data. **1**

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Question 9 (2 marks)

Given $\log_c(y) = \log_c(A) + tk \log_c(b)$, where A, b, c, k are (constant) positive real values with $b \neq 1$ and $c \neq 1$, and t and y are variables, show that $y = Ab^{kt}$.

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Examination continues on next page

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.

In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

Solve $3^x = 25$, giving your answer to 2 significant figures.

2

$$x = \log_3(25)$$

① mark

$$x = 2.9$$

① mark (2 sig. fig.)

Question 2 (2 marks)

Solve $3 \log_2(10) + \log_2(x) = 6$. Express answer in simplest form.

2

$$\log_2(10^3) + \log_2(x) = 6$$

$$\log_2(1000x) = 6$$

$$1000x = 2^6$$

$$x = \frac{2^6}{1000}$$

$$x = \frac{8}{125}$$

① mark: correctly simplifying surds

① mark: fully simplified

Question 3 (3 marks)

Solve $2^{x^2} = 8^{\frac{2}{3}x+1}$.

3

$$2^{x^2} = (2^3)^{\frac{2}{3}x+1}$$

$$2^{x^2} = 2^{2x+3}$$

$$\Rightarrow x^2 = 2x+3$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

① mark: correctly converting to same base

① mark: recognising quadratic

① mark: correct solutions

Examination continues on next page

Question 4 (3 marks)

Differentiate the following, simplifying where necessary.

a) $y = 2e^{4x-3}$

1

$$\frac{dy}{dx} = 8e^{4x-3}$$

① mark

b) $y = (2 - e^{-x^2})^6$

2

$$\frac{dy}{dx} = 6(2 - e^{-x^2})^5 \times 2xe^{-x^2}$$

① mark

$$= 12xe^{-x^2}(2 - e^{-x^2})^5$$

① mark: fully simplified

Question 5 (2 marks)

If $\log_x(6) = a$ and $\log_x(3) = b$, find $\log_x(8)$ in terms of a and b .

2

$$\log_x(8) = \log_x(2^3)$$

$$= 3\log_x(2)$$

$$= 3[\log_x(6) - \log_x(3)]$$

① mark: recognising 8 is a power of 2 and simplifying

$$= 3(a - b)$$

① mark: correct answer

Question 6 (5 marks)

Radioactive decay is a process which causes a substance to break down over time. The half-life of a substance refers to the amount of time it takes for the substance to decay to half its original mass.

Consider the formula for the radioactive decay of the substance carbon-14,

$$C = C_0 e^{-kt},$$

where C_0 and k are constants, t is time in years, and C is the amount of carbon-14 measured in grams.

- a) Find the value of C_0 if there is initially 1000 grams of carbon-14.

1

$$1000 = C_0 e^{-k \times 0}$$

$$C_0 = 1000$$

① mark: correct answer

- b) The half-life of carbon-14 is known to be 5730 years.

3

Use this to show that $k = \frac{\ln(2)}{5730}$.

$$500 = 1000 e^{-5730k}$$

$$e^{-5730k} = \frac{1}{2}$$

① mark: correct value of C and t

$$-5730k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{\ln\left(\frac{1}{2}\right)}{-5730}$$

① mark: correctly solving for k

$$k = \frac{\ln(2^{-1})}{-5730}$$

$$k = \frac{\ln(2)}{5730}, \text{ as req.}$$

① mark: recognising $\frac{1}{2} = 2^{-1}$ and simplifying

- c) Hence, find the amount of carbon-14, to the nearest gram, after 10,000 years.

1

$$C = 1000 \times e^{-\frac{\ln(2)}{5730} \times 10,000}$$

$$= 298g$$

① mark: answer, correctly rounding

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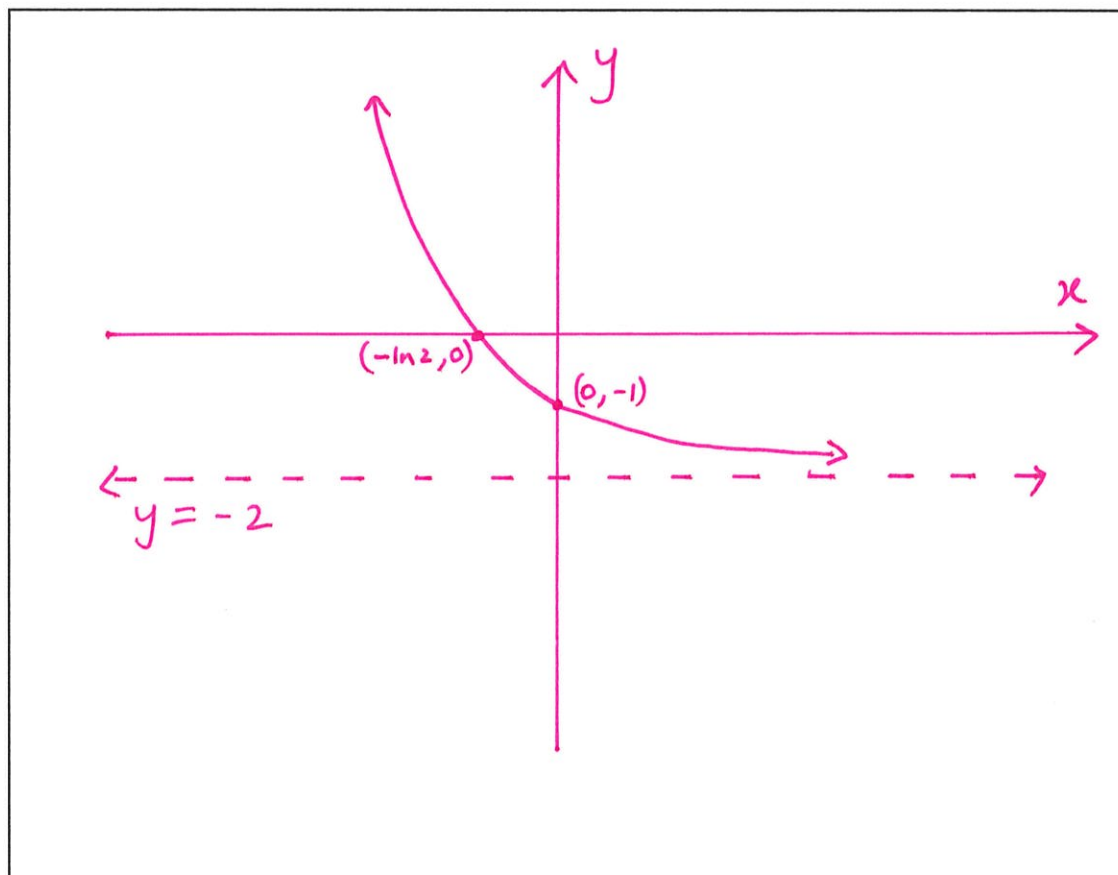
Question 7 (3 marks)

- (a) Sketch the graph of the following exponential function in the box below.
Label x and y intercepts and the asymptote.

3

$$y = e^{-x} - 2$$

Note: The box suggests the size of the expected graph



Asymptote: as $x \rightarrow \infty$, $y \rightarrow -2 \Rightarrow y = -2$ is the asymptote

① mark: correct asymptote and correctly drawn with a dotted line and labelled

x -int: set $y = 0$; $e^{-x} - 2 = 0$
 $e^{-x} = 2$

$$-x = \ln(2)$$

$$x = -\ln(2)$$

y -int: set $x = 0$: $y = e^0 - 2$
 $= -1$

① mark: correct intercepts ~~and~~ and correctly labelled

① mark: correct ~~y-int~~ and correctly labelled shape of curve

Question 8 (6 marks)

Sound intensity level (L) measured in decibels (dB) is a logarithmic measure given as:

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

The reference level of the intensity of sound (I_0) is 10^{-12} watts/m².

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Cello	87	(i)
Violin	93	0.0019953
Trombone	100	0.0100000
Timpani and bass drum rolls	(ii)	0.0501187

Determine the missing values of (i) and (ii).

i) $87 = 10 \log_{10} \left(\frac{I}{I_0} \right)$

$\log_{10} \left(\frac{I}{I_0} \right) = \frac{87}{10}$

$\frac{I}{I_0} = 10^{8.7}$

$I = 10^{8.7} \times 10^{-12}$

$= 0.0005012$

correct working and

① mark: correct to 7 decimals

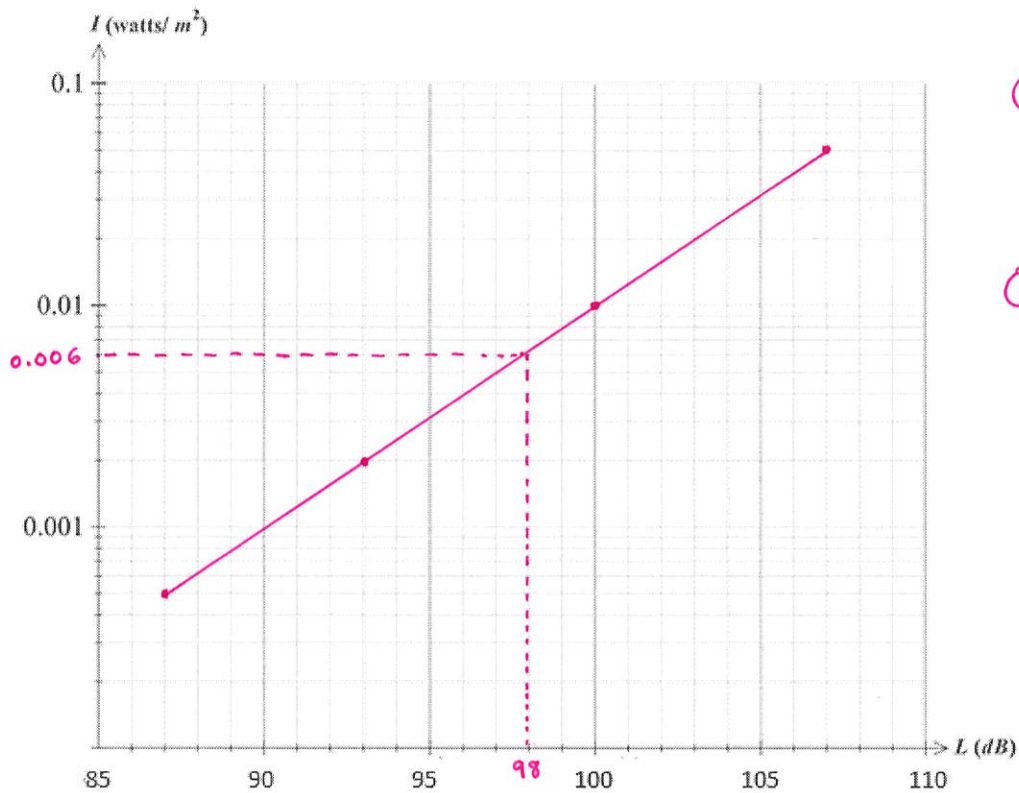
ii) $L = 10 \log_{10} \left(\frac{0.0501187}{10^{-12}} \right)$

$= 107$

① mark: correct to nearest dB

Examination continues on next page

- (b) Plot the intensity of sound (I) against the average sound intensity levels (L) for the **four instruments listed in part (a)** on the semi-logarithm graph below. 2
Connect the points with a line.



① mark:
correctly plotted
points

① mark:
lines drawn
through the
points

- (c) Using the graph determine the average sound intensity level (dB) of the sound from a musical instrument that has an intensity of 0.006 watts/m². 1

98 dB

① mark: correct answer
with dotted lines
shown on graph

- (d) Explain why a semi-logarithm grid is useful for this data. 1

It makes it easier to read the values from the graph compared to a linear scale on both axes, especially for very small or large values.

① mark: equivalent
logical and sensible
explanation

Question 9 (2 marks)

Given $\log_c(y) = \log_c(A) + tk \log_c(b)$, where A, b, c, k are (constant) positive real values with $b \neq 1$ and $c \neq 1$, and t and y are variables, show that $y = Ab^{kt}$.

2

$$\log_c(y) = \log_c(A) + \log_c(b^{kt})$$

$$\log_c(y) = \log_c(Ab^{kt})$$

① mark: correctly applies log laws

By taking exponential of both sides with base $c \Rightarrow y = Ab^{kt}$, as req.

① mark: correct answer with logical working

Examination continues on next page